WEMAKEDEVS × AWS ENVIRONMENTAL HACKS · WASTE & ENERGY TRACK

Lane Scorecard
Project overview

A campus tool that turns quick checks of NSUT's green and blue bins into specific feedback, gives cleaning staff a way to report without reading, puts their safety first, and checks honestly whether anything improved.

Live: **https://ss9ib2qbek.execute-api.us-east-1.amazonaws.com**
Team Galgotia Robots · Version 1 · 10 October 2026
Labels used in this document: Real measured or built and tested · Simulated demo data · Not yet proven a claim we cannot make yet.

1. The problem

Across NSUT, bins stand in pairs: green **“Degradable waste (kitchen waste)”** and blue **“Non-degradable waste (recyclable)”**. Having two bins does not mean waste is separated, and today nobody can tell whether it is.

- **No record.** Bins are not checked regularly, so nobody knows how bad it is or where.
- **No feedback.** Posters say “please segregate”; nobody tells people what exactly they got wrong.
- **The people who suffer have no voice.** Cleaning staff sort mixed waste by hand, sometimes with broken glass, blades or needles inside.
- **Sorting can be undone at pickup** if both bins go into one cart, and nobody notices.
- **Fixes are not checked.** Something is tried, then forgotten.

What we saw on our own campus Real Ratings not yet confirmed on site

On the night of 9 October we photographed bin pairs across campus (50 photos) and entered 9 checks from them into the live real pilot:

9	7 of 7	7 of 7	6
checks at 9 bin spots, one of them the canteen	green bins we could see inside had a mistake (2 “a few wrong”, 5 “mixed”)	blue bins we could see inside looked correct	checks tagged “plastic in the green bin”

The pattern: **the green bin is being used as a general bin** — wrappers, cups, bottles, a plastic bag — while the blue bin is fine. The canteen puts food and plastic together into black bags and onto the tray counter. Two green bins were too dark or empty to rate and are recorded as “Not seen”, not as clean.

> These ratings were read from photos and still need confirming on site; the place names are placeholders until renamed. One night of photos is a baseline, not a trend.

2. Our thought process — how the project got here

Step	What we learned → what we changed
Start: hostel scorecard	Two bins per floor, a weekly score and a private card per floor. Recognition, not money; no names.
Looked at rival entries	Most Waste & Energy projects give sorting advice or points. Nobody checks whether a fix worked, and nobody designs for the sanitation worker. → We made those two things the core.
"Did it work?" honestly	Simulated data showed weekly scores swing about ±30 points by chance with one check a day. A naive "down 20 points" would mislead. → The app only says *clearly better / worse* when the change beats normal ups and downs; otherwise "no clear change". A code review then showed our first version compared two different averages; we fixed it so the change and the noise use the same numbers.
Thought as the people involved	Cleaning staff may not read English, or read at all. → A picture-only page with Hindi voice and a red danger button; dangerous items jump to the top of the plan. Students ignore generic tips. → Specific four-part feedback from "what was wrong" tags.
Reality check: our hostel	One big mixed bin per corner and no personal bins, so there is nothing to rate as "separated". → We moved to the paired bins across campus.
Whole NSUT campus	Locations get areas (Academic blocks, Hostels, Canteen & food…), up to 200 bin pairs, NSUT's own bin labels, and a per-area summary.
Real photos	50 night photos sorted by place; a Photos page to review and delete wrong photos; thumbnails in the log; filter by location and day.
Looks like a real tool	Plain page names instead of slogans, sign-out always visible, simpler sign-in.

Principles we kept throughout

- **People decide ratings.** AI photo hints exist but are off; a person always makes the call.
- **Missing is never clean.** A bin nobody saw is "Not seen" and never raises a score.
- **No names, no rankings, no points.** Locations, not people; recognition, not money.
- **Say what we cannot prove.** Simulated data is labelled everywhere; nothing claims waste went down.

3. How it makes an impact

Our theory of change, step by step, with what is already true and what still has to be shown:

If…	…then	Status
Bins are checked regularly	There is a record of where and how separation fails, by place and campus area.	Real 9 checks entered
Feedback names the exact mistake	People know what to change (“tip the food out, put the bag in the blue bin”), instead of a generic poster.	Built effect not yet measured
Cleaning staff can report in 4 taps	The people who see every bin add daily data, and their safety becomes the top action.	Built not yet tried with staff
Mixing at pickup is recorded	The estate office sees when sorting is undone by collection, and a fix is only closed after 3 separate clean days.	Built
Next week compares honestly	The team learns whether an action worked, without fooling itself with normal ups and downs.	Built needs 2+ weeks of data

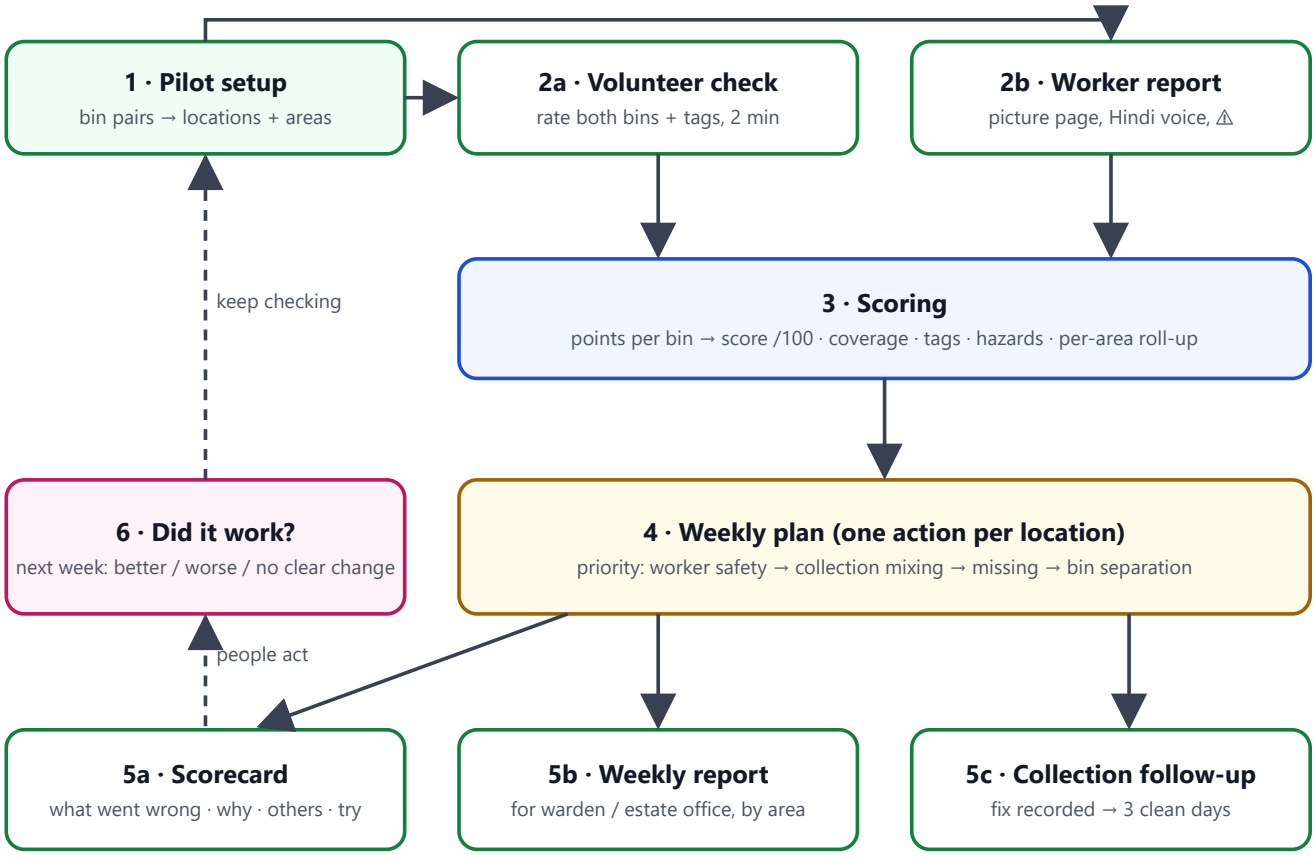
What would show real impact

- Two or more weeks of checks at the same places, so “vs last week” can say clearly better or worse.
- Fewer “plastic in the green bin” tags after the cards are shared at those places.
- Cleaning staff reporting on their own, and fewer dangerous items flagged over time.
- The estate office acting on the per-area report.

What we do not claim: that waste went down, that cards changed behaviour, that AI rates bins, or that staff have used it. One night of photos is our baseline.

4. The complete workflow

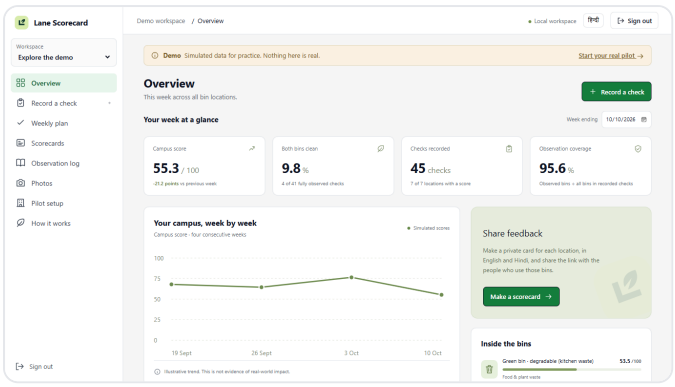
One loop that repeats every week. Each step names the screen in the app where it happens.



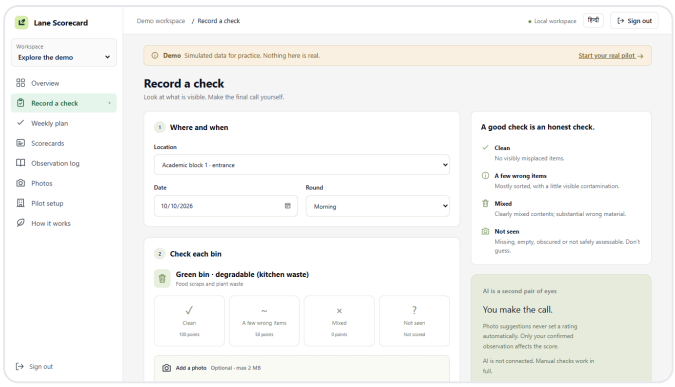
Solid arrows: data flow inside the app. Dashed: what people do between weeks.

Step	Who	What happens · where
1 · Pilot setup	Pilot lead	Add each green/blue pair as a location with its area; create the collection staff link and send it on WhatsApp · *Pilot setup*
2a · Volunteer check	Team / warden	Rate each bin Clean / A few wrong / Mixed / Not seen, tick what was wrong, note pickup, add a photo · *Record a check*
2b · Worker report	Cleaning staff	Tap place → faces for each bin → optional picture tags → red danger button → Send; spoken in Hindi · *worker link*
3 · Scoring	App	Clean 100 · few 50 · mixed 0 · Not seen excluded; score, coverage, tags, hazards, per-area summary · *Overview*
4 · Weekly plan	App → pilot lead	One action per location, in priority order, with the evidence · *Weekly plan*
5a/b/c · Share & act	Everyone	Four-part cards for residents (*Scorecards*); a printable report by area (*Download weekly report*); collection fixes tracked to 3 clean days (*Weekly plan*)
6 · Did it work?	App	Next week each location says clearly better, clearly worse, or no clear change · *Weekly plan, report*

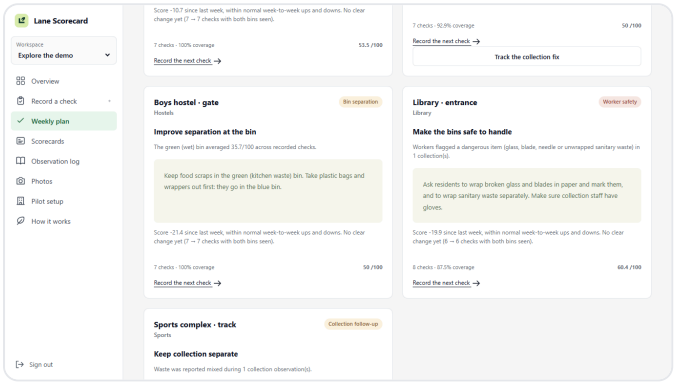
5. What the screens look like



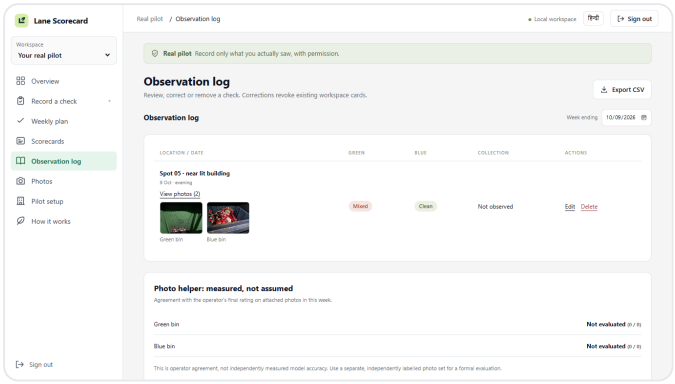
Overview — this week across all locations (demo)



Record a check — two minutes per bin pair



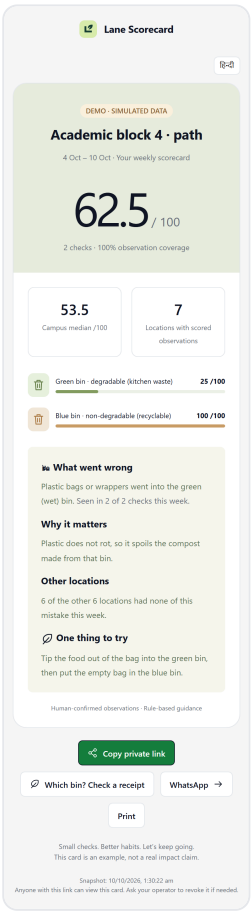
Weekly plan — worker safety comes first (demo)



Observation log — real photos from 9 Oct as thumbnails

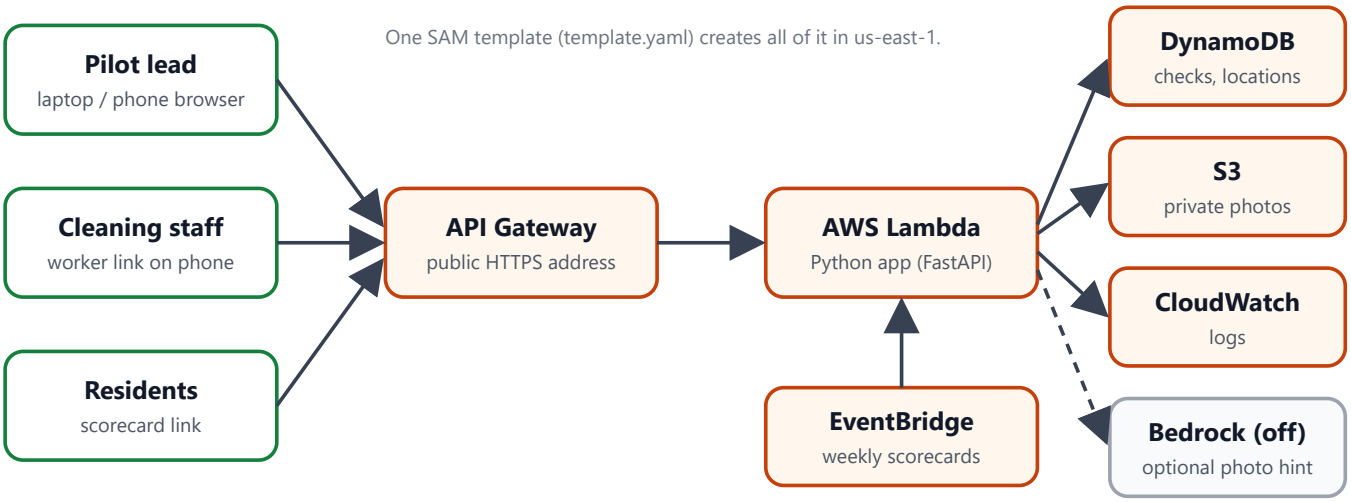


Worker page — no reading needed, Hindi voice



Scorecard — what went wrong, why, others, one thing to try

6. How it is built



Part	What it is
App	Python (FastAPI) server + plain JavaScript pages; English and Hindi; works on phones.
AWS	API Gateway, Lambda, DynamoDB, S3, EventBridge and CloudWatch from one SAM template, deployed in us-east-1. Bedrock photo hints built but switched off.
Privacy & safety	Sign-in for the pilot lead; secret, revocable worker and card links; no names; notes and photos never on cards or reports; rate limits; metadata stripped from photos.
Testing	90 automatic tests, plus real-browser checks of the warden app, collection follow-ups, the worker page on a phone, photo filters and delete.
AI tools used	OpenAI Codex (most of the first implementation, collection follow-ups); Claude / Claude Code (research, planning, worker page, follow-up check, campus scale-up, photos, AWS deployment, docs).

7. Where we stand and what is next

Done	Next
• App live on AWS, campus-wide • 9 real checks + 14 photos from 9 Oct • Worker page, follow-up check, photos tools • Team guide, testing guide, user guide PDFs	• Rename the “Spot” places to real NSUT names and confirm ratings • Two more rounds of checks at the same places • Try the worker page with 1–2 cleaning staff, with consent • Record the 3-minute video showing AWS; submit by Sunday 4 PM • Keep the organisers’ written answer on our early start with the submission